



Programming Fundamentals

Conditional Statements

Programming Fundamentals Team

SETU

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Outline

1. Flow of Control in a Program
2. Boolean Conditions
3. Relational Operators
4. Conditional Statement - if statement



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1. Flow of Control in a Program

Each program you write will typically have:



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Sequence	Things that will be done in a particular order
Selection	Things that will be done conditionally
Iteration	Things that will be done repetitively



1. Flow of Control in a Program

For this slide deck, we will be covering the first two types of control flow: sequence and selection. We will cover iteration in a later slide deck.



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1. Flow of Control in a Program

For this slide deck, we will be covering the first two types of control flow: sequence and selection. We will cover iteration in a later slide deck.

Sequence	Things that will be done in a particular order
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Sequence is the default flow of control in a program. It simply means that the statements in a program will be executed in the order they are written.



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Sequence	Things that will be done in a particular order
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- For the rest of this slide deck, we will be covering selection statements.
- Selection statements allow us to execute certain statements only if a certain condition is true.
- The most common selection statement is the if statement.



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2.1 Data types - Boolean

```
test10.py > ...
1  isOpen = True
2  isClosed = False
3
4  print('Is the shop open? ' + str(isOpen))
5  print('Is the shop closed? ' + str(isClosed))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
maireadmeagher@Mac dev % /opt/homebrew/bin/python3 /Users/maireadmeagher/Docu
Is the shop open? True
Is the shop closed? False
maireadmeagher@Mac dev %
```

- Remember our discussion of data types? We will be using the data type called **Boolean** in selection statements.
- A **Boolean** is a data type that can only have two values: True or False.



2.1 Data types - Boolean

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```

- Remember our discussion of data types? We will be using the data type called **Boolean** in selection statements.
- A **Boolean** is a data type that can only have two values: True or False.
- Booleans are often used in programming to represent the truth value of a condition or to control the flow of a program based on certain conditions.



2.2 Boolean Conditions

- A Boolean condition is an expression that evaluates to either **True** or **False** e.g.

```
x < 50  
number > 0
```



2.2 Boolean Conditions

- A Boolean condition is an expression that evaluates to either **True** or **False** e.g.

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- A conditional statement evaluates a Boolean condition and its result will determine which section of the code is executed.



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3. Relational Operators

Operator	Syntax	Returns true if...
>	x > y	x is greater than y
>=	x >= y	x is greater than or equal to y
<	x < y	x is less than y
<=	x <= y	x is less than or equal to y
==	x == y	x and y are equal
!=	x != y	x and y are not equal



3. Relational Operators

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==	x == y	x and y are equal
!=	x != y	x and y are not equal

= is an assignment operator and doesn't test for equality.
== is used to test for equality.



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4.1 if statement syntax

`if` *perform some boolean test* :

Do these statement(s) if the test is true

← actions if
condition is true



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NOTE:

- Python relies on indentation to define scope in the code.
- Indentation is a whitespace at the beginning of the line.
- We saw this earlier when writing functions.



4.1 if statement syntax

Example of an if statement:

```
age = 25
if age >= 18:
    print("You are an adult.")
    print("You can vote.")
```



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Example of an if statement:

```
age = 25
if age >= 18:
    print("You are an adult.")
    print("You can vote.")
```

This will output:

```
You are an adult.
You can vote.
```



4.1 if statement syntax

Example of a sequence of if statements:

```
age = 18
if age > 18:
    print("You are an adult.")
    print("And you are over eighteen")

if age == 18:
    print("Congratulations on reaching
adulthood!")

if age < 18:
    print("You are a minor.")
```



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Example of a sequence of if statements:

```
age = 18
if age > 18:
    print("You are an adult.")
    print("And you are over eighteen")

if age == 18:
    print("Congratulations on reaching
adulthood!")

if age < 18:
    print("You are a minor.")
```

This will output:

```
Congratulations on reaching adulthood!
```



4.1 if statement syntax

==if-else statement syntax